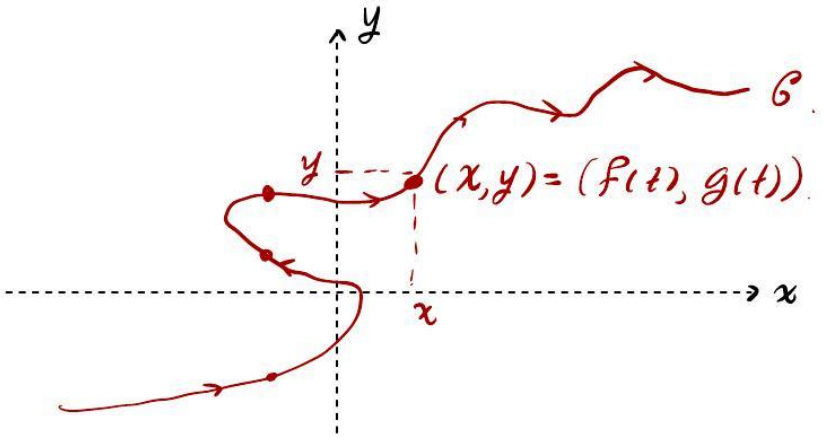
Lecture 17, Mar 06 2026

Imagine that a particle moving along the curve C



In general C is not a function $y = f(x)$. But

Parametric equations.

Suppose that x and y are both given as functions of the variable t, called a parametric equations by the equation

$$x = f(t) \qquad y = g(t)$$

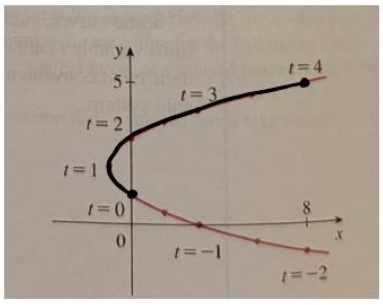
Each value t we have $(x,y) = (f(t), g(t))$ the curve is called a parametric curve.

Example. sketch and identify the curve defined by.

$$x = t^2 - 2t \;,\; y = t + 1$$

Sln.

$t = -2$	$t = -1$	$t = 0$	$t = 1$	$t = 2$
$x = 8$	$x = 3$	$x = 0$	$x = -1$	$x = 0$
$y = -1$	$y = 0$	$y = 1$	$y = 2$	$y = 3$



Now if $0 \le t \le 4$. then if $t=0$ $(0,1)$ - initial point.

if $t=4$ $(8,5)$ - terminal point.

Rmk. In general if

$$x=f(t) \quad y=g(t) \quad a \le t \le b$$

then $(f(a), g(a))$ is the initial point

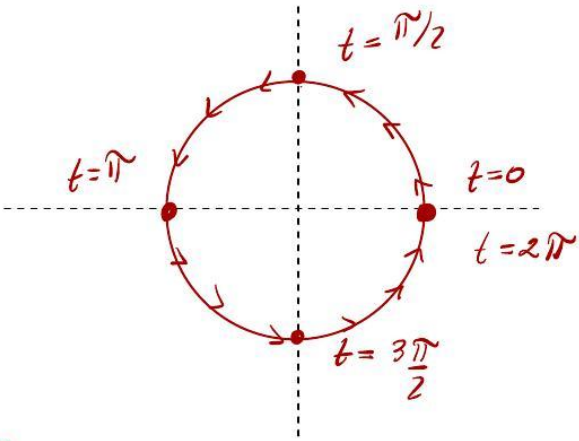
$(f(b), g(b))$ is the terminal point.

Example. What curve is represented by the equations?

$$x=\cos t \quad y=\sin t \quad 0 \le t \le 2\pi.$$

Sln: $t=0 \Rightarrow (1,0)$; $t=\frac{\pi}{2} \Rightarrow (0,1)$; $t=\pi \Rightarrow (-1,0)$; $t=\frac{3\pi}{2}$

$\Rightarrow (0,-1)$; $t=2\pi \Rightarrow (1\ 0)$



By eliminating the parameter t.

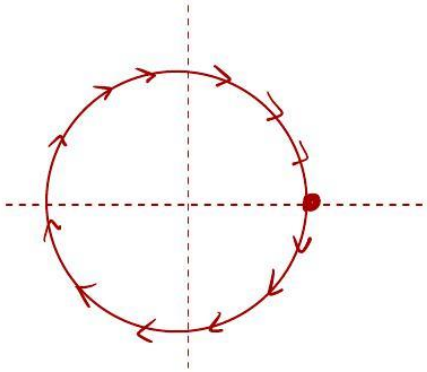
$$x^2+y^2=\cos^2 t+\sin^2 t=1.$$

then

$$x^2+y^2=1.$$

For example: $x=\sin 2t$, $y=\cos 2t$ $0 \le t \le 2\pi$

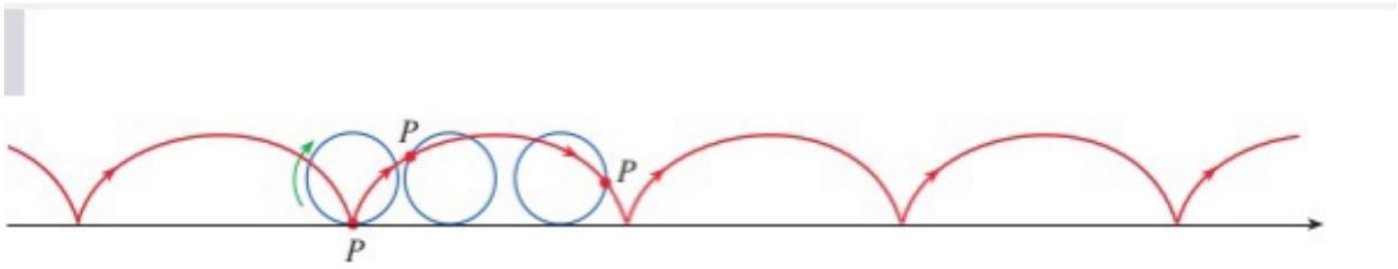
$$x^2+y^2=\sin^2 2t+\cos^2 2t=1.$$



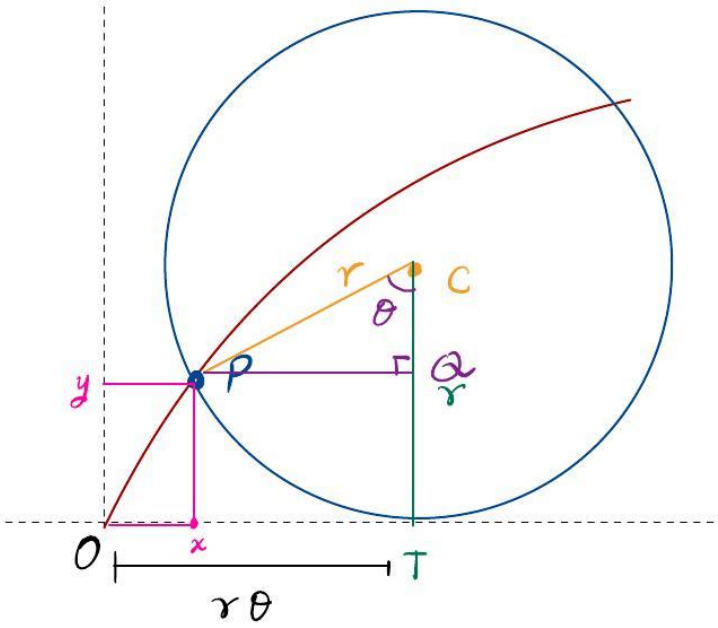
the particle moves twice around the circle in the clockwise direction.

The Cycloid.

Example The curve traced out by a point P on a circumference of a circle as the circle rolls along a straight line is called a cycloid. If the circle has radius r and rolls along the x-axis and if one position of P is the origen, find the parametric equations.



Sln.



Notice that

$|OT| = \text{arc } PT = \overset{\frown}{PT} = \theta \cdot r$

$x = |OT| - |PQ|$

$y = |TC| - |CQ|$

$|OT| = r\theta$

$|PQ| = r\sin\theta$ b/c.

$\sin\theta = \dfrac{|PQ|}{r}$

$|TC| = r.$

$|CQ| = r\cos\theta$ b/c

$\cos\theta = \dfrac{|CQ|}{r}.$

$$x = r\theta - r\sin\theta = r(\theta - \sin\theta)$$

$$y = r - r\cos\theta = r(1 - \cos\theta) \qquad \theta \in \mathbb{R}.$$

10.2 Calculus with parametric equations.

* Tangents.

Let f, g are differentiable functions and

$$x = f(t), \quad y = g(t)$$

the parametric equations.

$$\frac{dy}{dt} = \frac{dy}{dx} \cdot \frac{dx}{dt}$$

then

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$$

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{\frac{d}{dt}\left(\frac{dy}{dx}\right)}{\frac{dx}{dt}}$$

Example. a) Find the tangent to the cycloid $x = r(\theta - \sin\theta)$ $y = r(1 - \cos\theta)$ at $\theta = \pi/3$.

b) At what points is the tangent horizontal? When is it vertical?

Sln. The slope of the tangent is

$$\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta}$$

$$\frac{dy}{d\theta} = \frac{d}{d\theta}\left(r(1-\cos\theta)\right) = \frac{d}{d\theta}\left(r - r\cos\theta\right) = 0 - r\frac{d}{d\theta}(\cos\theta) = r\sin\theta$$

$$\frac{dx}{d\theta} = \frac{d}{d\theta}\left(r(\theta - \sin\theta)\right) = r(1 - \cos\theta)$$

$$\frac{dy}{dx} = \frac{r\sin\theta}{r(1-\cos\theta)} = \frac{\sin\theta}{1-\cos\theta}.$$

$\theta = \pi/3$

$$\frac{dy}{dx} = \frac{\sin(\pi/3)}{1-\cos\pi/3} = \frac{\sqrt{3}/2}{1-1/2} = \sqrt{3}.$$

$m = \sqrt{3}$

$$x = r\left(\frac{\pi}{3} - \sin\frac{\pi}{3}\right) = r\left(\frac{\pi}{3} - \frac{\sqrt{3}}{2}\right)$$

$$y = r\left(1 - \cos\frac{\pi}{3}\right) = \frac{r}{2}$$

$P\left(r\left(\frac{\pi}{3} - \frac{\sqrt{3}}{2}\right), \frac{r}{2}\right)$

$x_1 \qquad y_1$

$$y - y_1 = m(x - x_1)$$

$$y - \frac{r}{2} = \sqrt{3}\left(x - r\frac{\pi}{3} + r\frac{\sqrt{3}}{2}\right)$$

$$y = \sqrt{3}x - r\pi\frac{\sqrt{3}}{3} + \frac{3}{2}r + \frac{r}{2}.$$

$$y = \sqrt{3}x + r\left(2 - \pi\frac{\sqrt{3}}{3}\right).$$